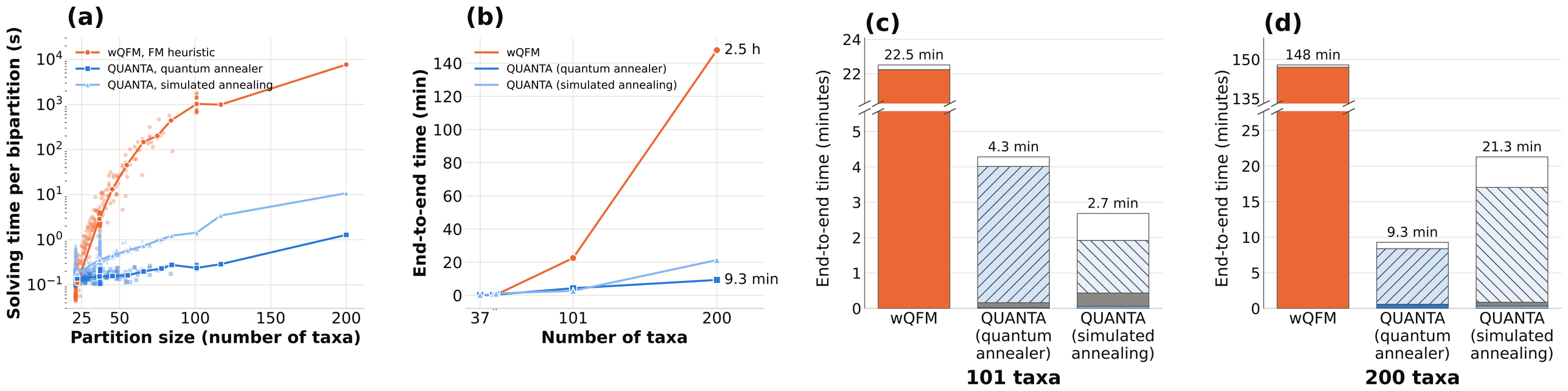

(a)
wQFM, FM heuristic
QUANTA, quantum annealer
QUANTA, simulated annealing
Solving time per bipartition (s)
Partition size (number of taxa)
(b)
wQFM
QUANTA (quantum annealer)
QUANTA (simulated annealing)
2.5 h
9.3 min
End-to-end time (min)
Number of taxa
(c)
22.5 min
4.3 min
2.7 min
End-to-end time (minutes)
wQFM
QUANTA (quantum annealer)
QUANTA (simulated annealing)
101 taxa
(d)
148 min
9.3 min
21.3 min
End-to-end time (minutes)
wQFM
QUANTA (quantum annealer)
QUANTA (simulated annealing)
200 taxa
wQFM: core FM step
QUANTA: quantum computer time
QUANTA: simulated annealing time
Exact search (partitions ≤ 20 taxa)
QUANTA (annealer): QUBO building + D-Wave access
QUANTA (sim. annealing): QUBO building
Rest of the run (reading quartets, merging, etc.)